SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA14 COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO1: Apply lexical analysis for token specification and recognition using LEX tool. *(BL3)*

Assessment Tool Weightage: 40%

Total Marks:50

QUESTIONS: 5

Hours : 60 mts

Mode: In Class

Annexure A

ANALYTICAL PROBLEM SOLVING

88

Code of Conduct:

I B·Hemanjali [192525457] *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B·Hemanjali

Q. No	Problem Understanding (20)	Method Selection (20)	Solution Process (25)	Accuracy of Calculation (20)	Interpretation of Results (15)	Total Score (out of 100)
1	/4	/4	/4	/4	/4	
2	/4	/4	/4	/4	/4	
3	/4	/4	/4	/4	/4	
4	/4	/4	/4	/4	/4	
5	/4	/4	/4	/4	/4	
OVERALL RUBRICS VALUE						
	/4	/4	/4	/4	/4	
	/20	/20	/25	/20	/15	/100

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Compiler Design

Name : B. Hemanjali
Reg No : 192525457
Faculty : Dr. N. Deepa

1)

$P = (a * b) + (b * c) * 2$

$id_1 = (id_2 * id_3) + (id_3 * id_4) * 2$

a) P, a, b, b, c - identifier (5)

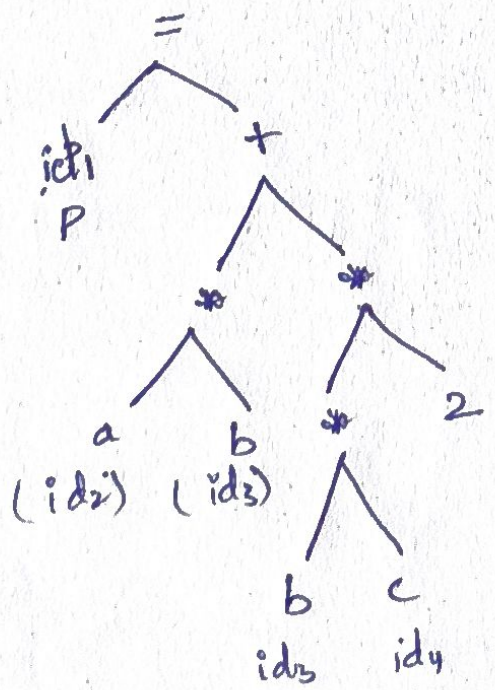
=, *, +, *, * - operators (5)

( ), ( ) - parentheses (4)

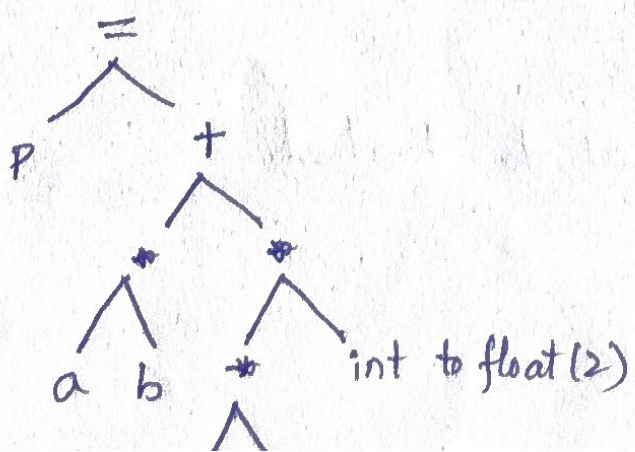
2 - integer (1)

Total Tokens = 15

b) Syntax tree



Semantic analysis



c) Intermediate Code Generator

$t_1 = a * b$

$t_2 = b * c$

$t_3 = t_2 * t_2$

$t_4 = t_1 + t_3$

$P = t_4$

d) Code optimization

$t_1 = a * b$

$t_2 = b * c$

$t_3 = t_2 * 2.0$

$P = t_1 + t_3$

e) Code Generation

```
LOADF  R1 , a
MULF   R1 , b
LOADF  R2 , b
MULF   R2 , c
MULF   R2 #20
ADDF   R1 , R2
STRF   P , R1
```

2) ① $(\varepsilon / aa^*)^*$

ε , a , aa , aaa , aaaa

② $(a^*/b^*)^* abb$

abb , aabb , babb , ababb , bbabb

③ $(a^*b)^* aba\ (a^* + b^*)^*$

aba , aaba , abaa , abab , babab

④ $(a/ab)^*$

ε , a , ab , aab , abab

$(aba)^* + (a^*b^*)^*$

ε, a, b, aba, bbaa

3) Lexical analysis

i) main . ( ) . { - 4

int , a , = , 10 , ; - 5

char , b , = , "abc" , ; - 5

in , t , c , = , 30 , ; - 6

ch , ar , d , = , "xyz" , ; - 6

in , t , m , = , 40.5 , ; - 6

} - 1

Total = 33 tokens

ii) lexical error

The lexical error is x = 1xab ;

∴ 1xab is not valid token . it starts with a digit so the scanner treats it as a number.

4) i) Even numbers of 0's

$1^*(01^*01^*)^*$

ii) at least one 1

$(0+1)^*1(0+1)^*$

iii) add numbers of 0's & 1's

$[00+11+(01+10)(00+11)^*(01+10)]^*(01+10)(00+11)^*$

iv) Start with 1 & end with 0

$1(0+1)^*0$

v) No consecutwe 1's

$(0+10)^*(\varepsilon+1)$

5) a) int var123 = 12345; float var456 = 123.45;

if (var123 == var456) { return }

Total tokens = 20.

b) Tokens by type

→ Keywords (4) : int, float, if, return.

* operators (3) : =, ==, ===
* Separators (7) : ;, ,, :, ( ) { ; }
* identifiers (4) : Var123, Var456, Var123, Var456
* integer literals (1) : 12345
* floating point literal (1) : 12345

c) 1) Keywords : int (3) + float (5) + if (2) + return (6)

Total = 16 transitions

2) Identifiers : Var123 & Var456, each have 3 letters & 3 digits, So 6 each. There are 4 identifiers

So 4 x 6 = 24 transitions

Total = 24 transitions

3) Integer literal : 12345 has 5 digits

Total = 5 transitions

4) Floating point : 123.45 has 3 digits & dot & 2 digits

Total = 6 transitions

5) operators : = , = , = =

Total = 4 transitions

6) Separators : three ; ( ), { }

Total = 7 transitions

Finally

Total Transitions = 16 + 24 + 5 + 6 + 4 + 7

= 62